AI4EO 2025 Symposium Assignment

CrossModMamba: Multi-Directional Memory Augmented Fusion of High-resolution SAR and Optical Imagery for Cropland Segmentation

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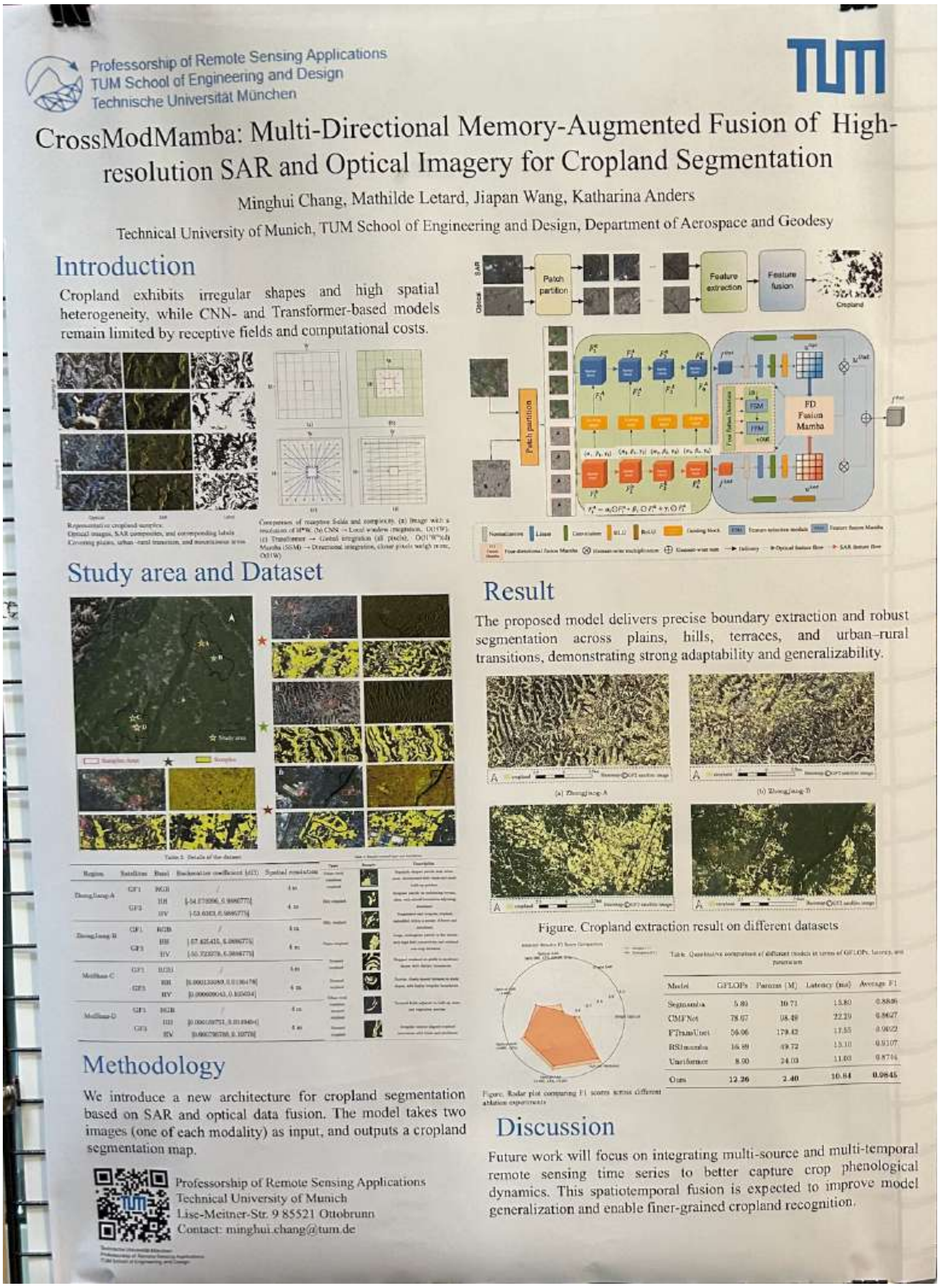

Figure 1: Selected Poster

Figure 1 represents the selected poster presentation from the AI4EO 2025 symposium.

1. Problem

The core problem addressed in this research is the accurate segmentation of cropland areas using high-resolution satellite imagery. Cropland fields often exhibit irregular shapes, fragmented boundaries, and high spatial heterogeneity due to factors such as terrain, soil type, crop type, and human activities. Traditional Convolution Neural Networks (CNNs) and transformer-based models struggle with these challenges. CNNs are limited by fixed receptive fields, making it hard to capture boundaries and details, while Transformer-based models require high computational costs due to quadratic complexity in processing high-resolution images.

2. Significance

With growing challenges from climate change and population growth, precise cropland mapping is a foundation for sustainable agriculture. Therefore, accurate cropland segmentation is critical for accurate crop type identification, crop yield estimation, environmental and land-use studies, disaster response, food security assessment, effective policy-making, such as resource allocation, and agricultural subsidies.

3. Strengths and Weaknesses

The three strengths of research are as follows:

a. The architecture is based on multi-modal data fusion that combines both Synthetic Aperture Radar (SAR) and optical data to overcome the limitations of each.
b. The proposed model extracts precise boundaries and robust segmentation, ensuring model adaptability and generalizability across plains, hills, terraces, and urban-rural transitions. This is because "CrossModMamba" is based on the incorporation of multi-directional memory that outperforms traditional CNN and Transformer-based models. The CNN captures local window features, the Transformer performs global integration across all pixels, and Mamba's State Space Models (SSM) integrate multi-directional memory augmentation techniques, giving more weight to closer pixels for precise contextual understanding. This helps with better handling of irregular cropland boundaries.
c. The proposed model achieves high performance with a remarkable inference time (10.4 ms), making the model suitable for real-time and large-scale applications.

The three weaknesses of research are as follows:.

a. The model is trained and evaluated on Chinese Gaofen-1 **(**GF-1) (Optical) and GF-3 (SAR) high-resolution (4 m) datasets, which are not openly accessible. The model generalization on the other sensors, such as Sentinel, Landsat, etc, that are in coarser resolution, are not fully explored.
b. The model is trained only on four regions of China (Zhong Jiang-A, Zhong Jiang-B, MeiShan-C, MeiShan-D), which leads to limited spatial generalization. This may

reduce the effectiveness of the model when applied elsewhere in China or other countries with different crop types, environmental, and geographical conditions.

c. Although optimized, the model still has a relatively high computational cost (12.26 GFLOPs) as compared to Segmamba (5.89 GFLOPs) and Unetformer (8.90 GFLOPs).

4. Extension/reuse of the Work in a Master's Thesis Topic

The following extension could be done as a master's thesis:

a. After the segmentation result, the work can be further extended to multi-temporal analysis by incorporating time-series SAR and optical data to capture crop phenology.

b. The model can be trained and tested in openly accessible datasets such as Sentinel and Landsat for crop segmentation, classification, and monitoring by collecting the training data from across the globe to ensure the model is generalizable both in terms of diverse spatial and environmental conditions. By improving model generalizability and scalability, the work can contribute effectively to the United Nations Sustainable Development Goals (SDGs), particularly Goal 2 (Zero Hunger).

c. The model achieves a remarkable average F1 (0.98) as compared to other cropland segmentation models, such as Segmamba (0.88), RS3mamba (0.91), Unetformer (0.87), etc., but can be optimized further to reduce the computational cost and inference time.

5. Explanation of Research to a General Audience

Imagine if the task is to delineate the boundaries of each cropland using drone or satellite imagery. As we know, croplands are rarely near rectangles, as most of them are irregular and vary in size. Traditional models, such as CNN or transformers, struggle to segment such non-uniform boundaries because CNN only captures the local window features at a time, whereas transformers consider all the pixels and are very computationally expensive to process high-resolution images. Therefore, this research introduces "CrossModMamba", an effective and efficient intelligence model to accurately segment the croplands.

This model combines two data modalities, i.e, Optical and SAR. Optical images capture colors and textures, whereas SAR can penetrate through clouds, work day/night, and can capture the surface structure and roughness. Each of these data types has its own strengths and weaknesses. Optical imagery shows fine details like the color of crops, but it is mostly covered by clouds. Radar imagery is reliable in all weather conditions and can capture surface textures, but it is harder to interpret visually. By combining both, models can benefits from each modality and can predict much sharper and more accurate boundaries of cropland.

In addition, the model acts like a multi-directional memory system, thereby remembering details from both SAR images and optical images through a memory augmented mechanism. It is based on Mamba's State Space Models (SSM) architecture that emphasizes directional integration to efficiently capture both local details and global contexts. This makes the model very powerful, and it can generate the precise boundaries across various landscapes such as plains, hills, and terraces. As a result of the experiment, this model outperforms the baseline models, such as Segmamba, Unetformer, etc. This research is helpful to improve agricultural planning, food security, policy making, and ultimately support the Sustainable Development Goals.